# 4.2 Writing Equations in Point-Slope Form

Name _____ Period ____ Date _____

Algebra I 4.2 · Student sheet

**I can…**

- write an equation of a line given its slope and a point.
- write an equation of a line given two points.
- use point-slope form to write linear models.

**Vocabulary** · fill in the missing word

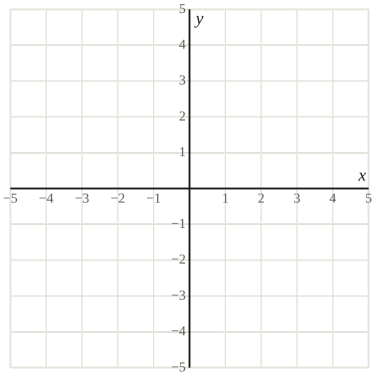
**Point-slope form** — $y - y_1 = m(x - x_1)$, for a line _____ $(x_1, y_1)$ with slope m

## Part 1 · Where does it come from?

Slope through $(x_1, y_1)$ and any point $(x, y)$: m = _____

Multiply both sides by $(x - x_1)$: _____

Use it: the line through (3, −1) with slope 2: _____




## Part 2 · Notes: fill in the blanks

Point-slope form: _____.

$(x_1, y_1)$ is a _____ on the line, and m is the _____.

$y + 2 = 3(x - 4)$ passes through _____.

## Part 3 · Examples (work them with the class)

### Example 1

Write an equation in point-slope form of the line through (4, −3) with slope −2.

### Example 2

Write an equation of the line through (−2, 5) and (1, −1) in point-slope form, then in slope-intercept form.

### Example 3

Is the table linear? If so, write an equation. x: 1, 3, 5, 7; y: 9, 15, 21, 27

### Example 4

A plumber charges a fixed fee plus an hourly rate. A 2-hour job costs $190 and a 5-hour job costs $385. Write an equation for the cost C of an h-hour job.

## Part 4 · Partner practice

### Partner 1

Write an equation in point-slope form through $(-5, 2)$ with slope $\frac{1}{2}$.

### Partner 2

f is linear, $f(2) = -1$, and $f(6) = 7$. Write f(x) in slope-intercept form.